GSSR-Net: Geo-Spatial Structural Refinement Network for Remote Sensing Change Detection

Shuo Wang, Genji Yuan, and Jinjiang Li

***Abstract*— Remote sensing (RS) change detection is a technique for identifying changes in the ground surface by comparing RS images from different periods. Although such tasks have been developed for a long time and some methods have been proposed to enhance the features of real changes in the ground objects, they still face the perception ambiguity caused by the heterogeneity of the ground objects in complex RS environments: 1) insufficient processing of nonstationary changes between dual-temporal image features and 2) high spatial heterogeneity leads to difficulties in structural identification. In order to solve the interference of these problems on the downstream tasks of change detection, this article proposes geo-spatial structural refinement network (GSSR-Net) for RS change detection. First, we introduce a DualTime Mamba structure with an omnidirectional scanning path, adjust its input matrix between dual-temporal image features in the deep scale, and allow the model to fully consider the spatiotemporal dependence and image structure information of the previous and next time points. In addition, this article designs a land-cover feature extraction (LCFE) method to improve the perception ability of the ground object target structure. Specifically, this method refines the image edge by separating high frequencies, adjusts the contour structure information of the dual-phase image by separating low frequencies, and then further models the relationship between pixels by combining feature structures and spatial offset mechanisms. Our experimental results on three datasets demonstrate the superiority of GSSR-Net. The network code address is https://github.com/SparrowTought/GSSR-Net.**

***Index Terms*— Building change detection (BCD), graph convolution, high-resolution remote sensing (RS) images, Mamba.**

I. Introduction

THE remote sensing (RS) change detection task aims to identify and analyze the changes in the Earth's surface by comparing RS images acquired at different times. This task is often used in the fields of environmental monitoring [1], urban development [2], [3], disaster assessment [4], land use change [5], [6], [7], etc. By comparing images at different time points, RS change detection can effectively reveal changes in surface cover types [8], the impact of natural disasters, urbanization progress [9], etc. In recent years, with the continuous development of deep learning technology, the application of convolutional neural networks (CNNs) in RS change detection has become a research hotspot [10], mainly due to its powerful feature extraction capabilities and end-to-end learning framework. Traditional RS change detection methods often rely on manually designed features and simple image difference analysis, while CNNs can automatically extract deep spatial features from multitemporal RS images, greatly improving the accuracy and robustness of change detection [11]. Deep learning models such as U-Net [12] and ResNet [13] have been introduced into RS change detection tasks, further improving the ability to capture image structure and identify changed areas [14], [15], [16].

In the RS change detection task, with the introduction of Transformer [17], the long-range dependence between different areas in the image can be effectively captured, overcoming the limitations of CNN in processing large-scale spatial relationships and ground structure. Unlike CNN, which mainly relies on local feature extraction, Transformer uses global modeling to enable the network to focus on broader structural information in RS images [18], [19]. Its advantage is that it can extract the global information of the image through the self-attention mechanism and further improve the recognition ability of changes in different time periods through multihead attention, which allows the model to better adapt to the complex spatial structure and multiscale changes in RS images [20], [21], [22]. Nowadays, the emergence of Mamba [23], [24], [25] inherits the advantages of Transformer through innovative architectural design and introduces new methods to improve computing efficiency, thereby significantly reducing the time required when processing long sequences or large-scale data. Computational overhead not only retains Transformer's ability to capture global information, but also significantly reduces the amount of calculation and improves efficiency through linear complexity design. The Mamba model is widely used in change detection tasks [26], [27]. It can efficiently process large-scale image data and significantly reduce the consumption of computing resources, showing wide application potential in big data or real-time analysis.

Although the above methods have achieved remarkable results in continuous innovation and have made breakthroughs in the processing of complex RS image structures, the traditional methods cannot deal with the heterogeneity of objects in complex RS images. This type of problem leads to two main problems: first, nonstationary changes make the dual-phase images more complex, and traditional methods are difficult to deal with the errors caused by seasonal,

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climate, or land use changes; and second, strong spatial heterogeneity leads to blurred object structures and uneven spatial distribution, which, in turn, affects the accuracy of change detection. In order to meet the challenges of such problems, this article proposes geo-spatial structural refinement network (GSSR-Net) for RS change detection. We introduce the DualTime Mamba coding structure, which captures the global structure of the image through a more comprehensive scanning mechanism, greatly reducing the computational overhead compared to the traditional VMamba model. This coding method learns the independent features of the current time point at a shallow scale and then interacts with the three matrices in the dual-phase state-space model (SSM) at a deep scale to capture the dynamic changes and complex spatiotemporal dependencies in the dual-phase image, thereby dealing with the nonstationary changes brought by the dual phases. We propose the land-cover feature extraction (LCFE) method to refine the structure of the ground object target. Specifically, the convolution operator in the spatial domain is combined with the high-frequency information in the frequency domain to refine the edge structure in the feature, and a normalization method for RS change detection tasks is proposed to balance the contour features and nonstationary changes in the low frequency. Then, the method uses graph convolution and local convolution to establish the regular and irregular structures of the features, and dynamically adjusts the relative position between the data of the two through the spatial offset field, thereby maintaining stable feature aggregation and improving the perception of the overall target structure of the RS image. The main contributions of this article can be summarized as follows.

1) In this article, a low-energy DualTime Mamba module suitable for dual-time RS change detection is designed to handle the nonstationary and complex spatiotemporal dependencies of changes in RS data by interacting with state matrices at different time points in the deep scale.
2) We designed the LCFE method, which optimizes edges and contours by separating high and low frequencies, and aligns regular and irregular structures using the offset field, thereby refining the overall structural nature of the features and improving the accuracy of downstream tasks.
3) We have demonstrated through a large number of experiments that the GSSR-Net proposed in this article is superior to the advanced methods. The experimental results show that GSSR-Net has obvious advantages in facing complex ground structure tasks.

The remainder of this article is organized as follows. Section II briefly reviews related work. The details of our proposed framework are described in Section III. Comprehensive experimental evaluations are conducted in Section IV. Finally, this article is concluded in Section V.

II. Related Work

A. Traditional CD Methods

Traditional RS change detection methods are mainly based on the analysis of image pixels, features, and object levels. Early methods such as the image difference method, principal component analysis (PCA), and change vector analysis (CVA) have laid the foundation for this field. The image difference method identifies the change area by comparing the pixel differences of images at different time points; PCA extracts the main features of the image through dimensionality reduction and analyzes the direction and intensity of the change; CVA combines spectral changes to detect the direction and magnitude of the change of the ground objects in the image. Singh [28] summarized these traditional methods and provided an important theoretical framework for RS change detection. Bruzzone and Serpico [29] proposed an iterative land-cover change detection method, which uses multitemporal images for pixel classification and improves the detection accuracy through iteration; Coppin and Bauer [30] summarized forest change detection technology and focused on the application of pixel-based change detection in forest ecosystems. Although modern technology has surpassed these traditional methods, they are still an important starting point for understanding the principles of change detection.

Algorithm 1 GSSR-Net

Input: X_1,X_2.*(bitemporal image)*
Output: O.
// Step1:Use the DualTime Mamba to fuse multi scale features in the encoder
for *i in 1, 2* **do**
  for *i in 1, 2, 3, 4* **do**
    $X_{i,j}$ = DualTime Mamba(X_i);
  end
end
// Step2: Use the LCFE module to Enhancement of the Physical Structure
for *i in 1, 2, 3, 4* **do**
  X_m = LCFE($X_{m,1}$, $X_{m,2}$);
  // Step3: Use the Diff-Enhance module to Feature Difference Fusion
  X'_m = Diff-Enhance($X_{m,1}$, $X_{m,2}$);
end
// Step4: Use the Decoder module
X_n = Decoder(X'_2, X'_3);
X'_n = Decoder(X'_1, X_n);
// Step5: obtain the result
O = Conv(X'_n)

With the advancement of technology, machine learning methods have gradually replaced traditional methods and become a new trend in RS change detection. Algorithms such as support vector machine (SVM) and random forest (RF) significantly improve change detection accuracy through data-driven analysis. Bruzzone and Prieto [31] proposed a change detection method based on SVM, which effectively utilizes high-dimensional feature space. Camps-Valls and Bruzzone [32] combined multiclass SVM to handle complex multiclass surface changes. Moser et al. [33] applied an RF algorithm to integrate learning to reduce the effects of noise and error. Although machine learning methods show strong

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flexibility and accuracy, they usually require large amounts of data and calculations, and need to be carefully selected and optimized for practical applications.

B. Deep Learning CD Method

CNN has become a common method in RS change detection. CNN automatically extracts features from images through layer-by-layer convolution and pooling operations without the need for manual feature design. The convolution layer can learn the spatial patterns in the image, while the pooling layer helps reduce the dimension, retain key features, and reduce computational complexity. Zhao et al. [34] proposed a multiscale feature-guided residual CNN method, which uses four different sizes of convolution kernels to extract multiscale features and fuse them to enhance representation capabilities. Zhang et al. [35] used CNN to detect candidate changes between two time periods, grouped these changes based on patches, and finally verified them as changes of a single object.

In RS change detection, the self-attention mechanism of Transformer can capture the dependencies between different pixels in the image and dynamically adjust their weights to help identify wide areas of change. Unlike CNN, Transformer does not rely on local convolution operations, but directly models global dependencies through the self-attention mechanism, making it particularly effective in processing long-sequence data and complex change patterns. Li et al. [36] proposed an end-to-end Transformer model for RS change detection, solving the problems of feature layer relationship modeling and feature difference representation. Zhu et al. [37] improved the Transformer structure and optimized the object detection performance in high-resolution images. Liu et al. [38] introduced a hierarchical Transformer structure and processed high-resolution images through a sliding window mechanism, which is suitable for computationally intensive visual tasks.

The Mamba model has significant advantages when processing long sequences and large-scale data. It inherits the global awareness capabilities of Transformer and improves computational efficiency through structural optimization. Especially in large-scale data processing, Mamba significantly reduces computational overhead through linear complexity algorithms, demonstrating its advantages in long-sequence data processing. Chen et al. [39] combined with the omnidirectional selective scanning module can model image context from multiple directions and capture a wider range of spatial features. Chen et al. [27] proposed three spatiotemporal relationship modeling mechanisms. When combined with the Mamba architecture, it can effectively realize the interaction of multitemporal features, thereby improving the accuracy of change detection.

III. Methodology

A. Framework Overview

In the framework overview, we will elaborate on the overall network architecture process and provide a comprehensive description of the specific methods of the DualTime Mamba and LCFE modules. As shown in Fig. 1, the specific framework is as follows.

First, GSSR-Net inputs the dual-phase feature image into the DualTime Mamba coding structure with weight sharing. After progressive extraction at four scales, the encoding obtains multiscale dual-phase images F1, F2, F3, and F4 with 64, 96, 192, and 384 channels. Second, the feature image of each scale is independently input into LCFE. After wave-edge sensing, the high- and low-frequency structures of the feature image are refined, and the refined edge and contour features are further stabilized to fuse regular and irregular structures. Subsequently, the network inputs the processed dual-phase features into the downstream task of the change detection operation and improves the accuracy of change detection through three methods: cosine similarity, absolute value difference, and channel superposition. Finally, the results of the four scales are decoded through deconvolution to increase the scale and channel superposition operations, and a multiweighted loss is set to deeply supervise the network structure. The network uses agrmax to output the final predicted image

B. DualTime Mamba

The traditional Mamba structure is mostly suitable for image segmentation and classification operations, but it is usually unable to handle the nonstationary changes faced by RS change detection tasks with dual-phase feature input. This situation is usually manifested as the amplitude, direction, or characteristics of the change being unstable over time or in the environment.

In this regard, this article proposes the DualTime Mamba structure to solve this problem. As shown in Fig. 2, we process the features of the four scales from shallow to deep in a progressive manner. The method introduces omnidirectional scanning: row scanning, column scanning, positive diagonal scanning, and negative diagonal scanning. The row scanning and column scanning create a relationship path on the boundary to ensure that the contextual semantic information can be more fully captured. We divide the depth of the four scales in DualTime Mamba into 2, 2, 9, and 2.

Specifically, we take the SSM as input $\varphi x(t) \in \mathbb{R}$, the hidden state as $\varphi h(t) \in \mathbb{R}^N$, and the output as $\varphi y(t) \in \mathbb{R}$, as shown in the following equation. Among them, the matrices $A \in \mathbb{R}^{N\times N}$, $B \in \mathbb{R}^{N\times 1}$, $C \in \mathbb{R}^{1\times N}$, and $D \in \mathbb{R}$

$$\begin{aligned} \varphi y(t) &= C\varphi h(t) + D\varphi x(t) \\ \varphi h(t) &= A\varphi h(t) + B\varphi x(t). \end{aligned} \tag{1}$$

The model discretizes the continuous system by introducing the S4 structure and effectively converts the continuous parameters A' and B' into discrete parameters and using the timescale parameter Δ, thereby maintaining the computational efficiency of the model when processing long-sequence data, as shown in the following equation:

$$\begin{aligned} A' &= \exp(\Delta A) \\ B' &= (\Delta A)^{-1}(\exp(\Delta A) - I)\Delta B. \end{aligned} \tag{2}$$

In the first two shallow scales, Mamba-Base is used to follow the processing of the current time $T_1 \in \mathbb{R}^{B\times H\times W\times C}$,

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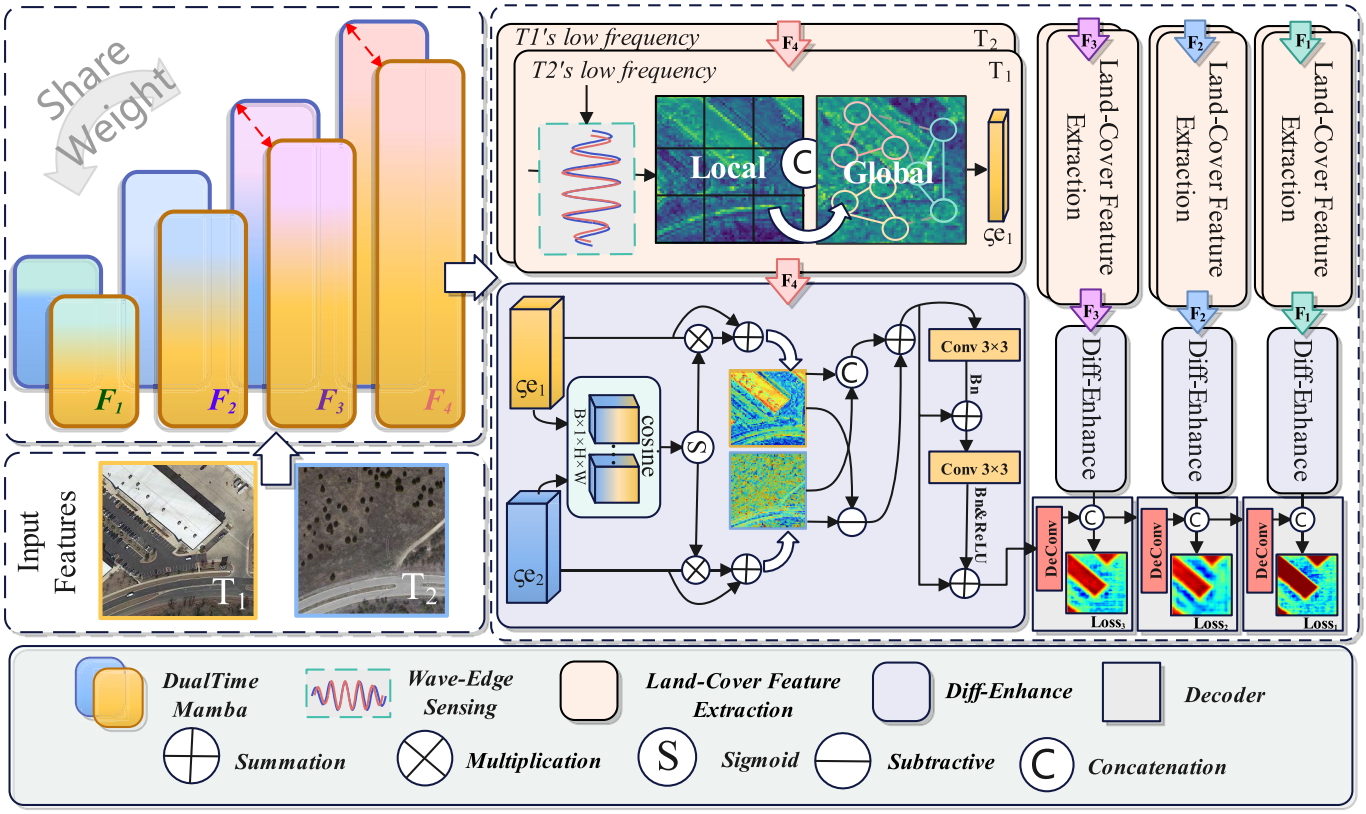


Fig. 1. Feature images at times T1 and T2 are input into the DualTime Mamba encoder, and four scales are output for parallel operation. Each scale is subjected to LCFE for feature structure perception, then the processed dual-time feature images are subjected to change detection through Diff, and finally, deconvolution is used for decoding operation.

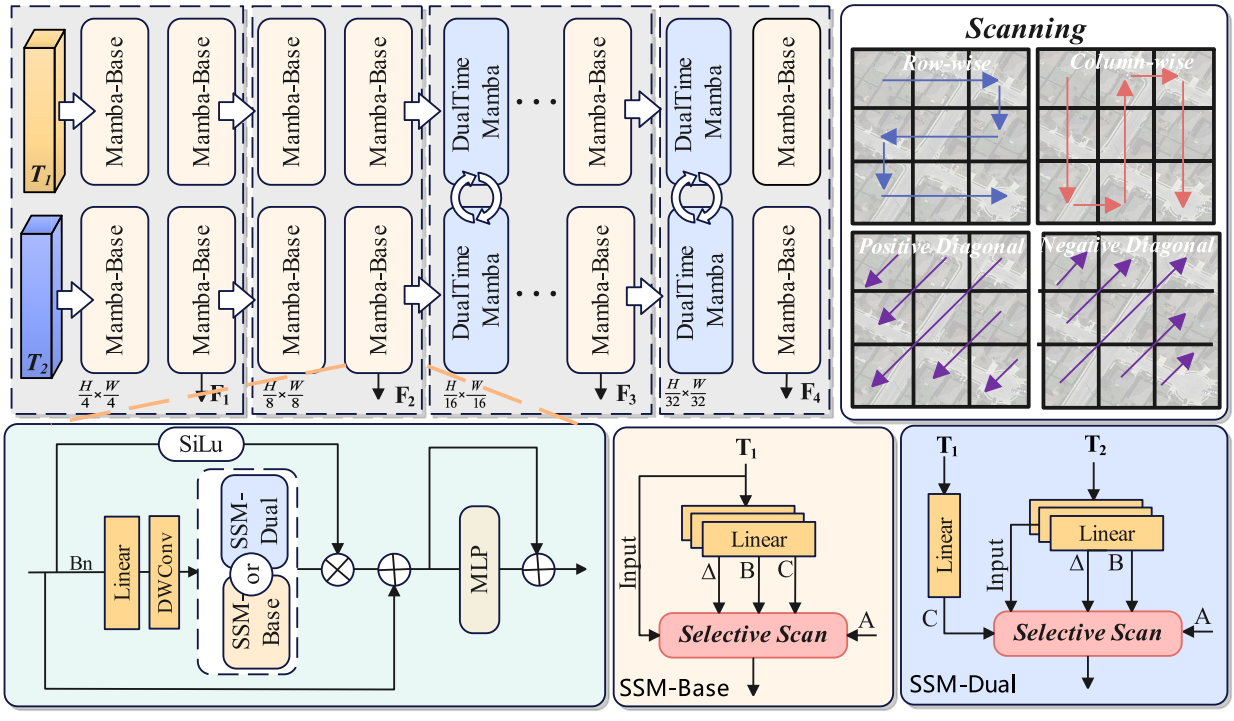


Fig. 2. DualTime Mamba uses an omnidirectional scanning strategy to implement the traditional SSM architecture in the first two scales and perform A, B, and D matrix interactions on the SSM in the last two scales, so that the model can dynamically adjust the mapping method of input features and control the fluctuation range of output features within a reasonable range, thereby filtering out interference caused by nonstationary changes such as seasonal fluctuations.

and the traditional SSM-Base structure is used to ensure that the network can learn the texture features of shallow features. Then, in the two deep scales, the SSM of the dual-phase features is interacted with the A, B, and D matrices, so that the three types of matrices at the current time depend on the feature information of another time $T_2 \in \mathbb{R}^{B\times H\times W\times C}$, which

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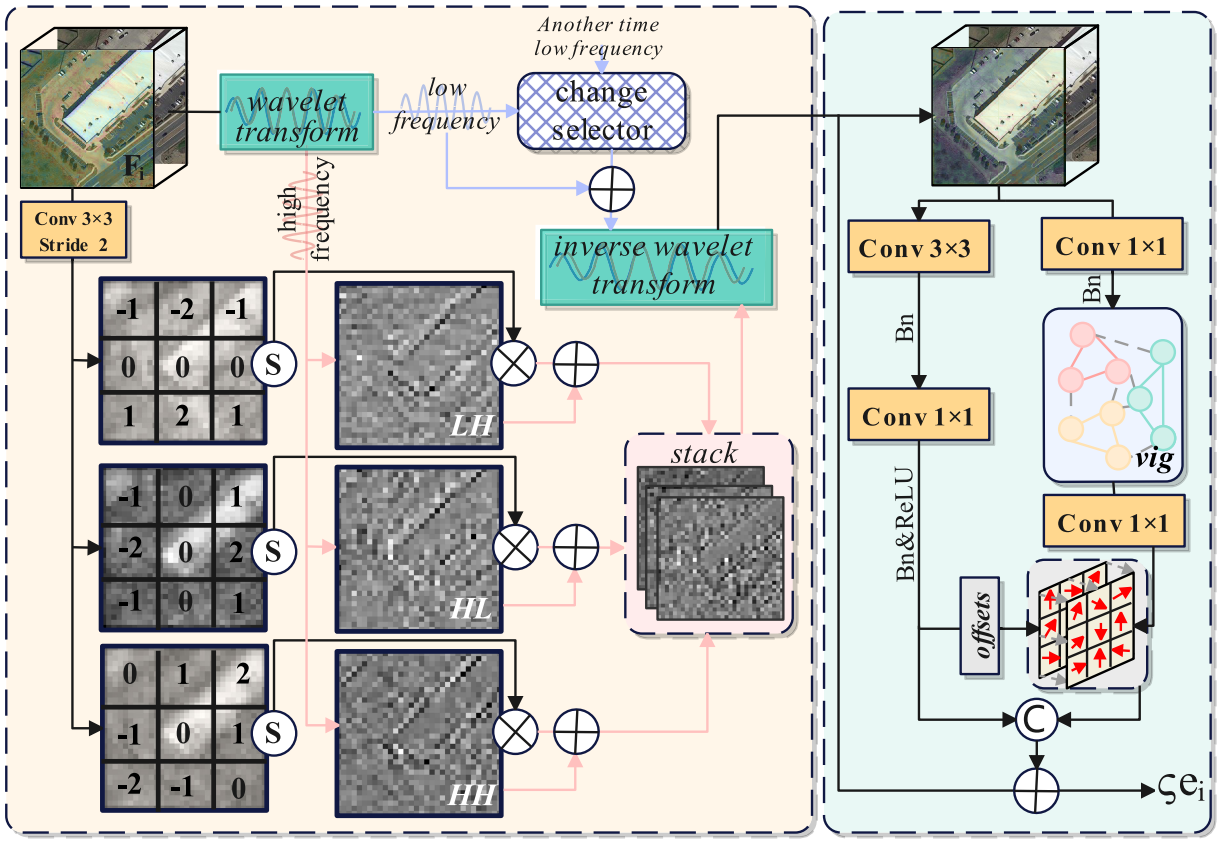


Fig. 3. LCFE separates high-frequency details and low-frequency contours through wavelet transform, combines the frequency-domain edges with the spatial-domain edges and normalizes the low-frequency contours by change selector, and then regularly fuses these structural information using graph convolution and local convolution, during which the pixels are aggregated and aligned through the offset field.

is currently shown in the following equation:

$$A..D = \text{Linear}_{A..D}(T_2), \quad \Delta = \text{Linear}_{\Delta}(T_2)$$
$$C = \text{Linear}_{c}(T_1). \tag{3}$$

This SSM-Dual structure allows the dynamic matrix to learn information in different time series while ensuring the static design of the matrix C, as shown in (4). The interaction of this structure can dynamically adjust the mapping method of the input features and control the fluctuation range of the output features within a reasonable range, thereby filtering out interference caused by nonstationary changes such as seasonal fluctuations

$$h_k = \bar{A}h_{k-1} + \bar{B}_{T_2}$$
$$y_k = Ch_k. \tag{4}$$

The same weight encoding structure is used for both temporal features. After SSM, nonlinear information integration is performed through multilayer perceptron (MLP), and the four scales are input into the subsequent tasks, where $i \in \{1, \ldots, 4\}$.

C. Land-Cover Feature Extraction

RS images usually have extremely complex edge structures and contour structure information, which often affect the change detection results due to unclear structures or light factors. In this regard, this article proposes an LCFE module, as shown in Fig. 3.

First, feature $\varphi F_i \in \mathbb{R}^{B\times C\times H\times W}$ is operated in parallel in the spatial domain and frequency domain. Specifically, downsampling is performed on the spatial branch through 3×3 convolution to correspond to the high-frequency scale after frequency separation in the frequency domain $\theta^f \in \mathbb{R}^{B\times C\times (H/2)\times (W/2)}$, and then, a convolution operator sensitive to gradient changes is used to establish the horizontal edge in the spatial domain θ^f_{HL}, the texture structure of vertical edge θ^f_{LH}, and oblique edge θ^f_{HH}, as shown in the following equation:

$$\theta^f_{LH} = \begin{pmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{pmatrix}, \quad \theta^f_{HL} = \begin{pmatrix} -1 & -2 & +1 \\ 0 & 0 & 0 \\ -1 & +2 & +1 \end{pmatrix}$$
$$\theta^f_{HH} = \begin{pmatrix} 0 & +1 & +2 \\ -1 & 0 & +1 \\ -2 & -1 & 0 \end{pmatrix}. \tag{5}$$

In the frequency-domain branch, the feature dimension is reduced to $\beta^f \in \mathbb{R}^{B\times C\times (H/2)\times (W/2)}$ by wavelet transform, and its high-frequency horizontal edge β^{high}_{LH}, vertical edge β^{high}_{HL} and oblique edge β^{high}_{HH}, and low-frequency β^{low} are separated, as shown in the following equation:

$$\beta^{\text{low}} = \sum_{k,j} \varphi F_i \cdot \phi(x-k) \cdot \phi(y-l)$$
$$\beta^{\text{high}}_{LH} = \sum_{k,j} \varphi F_i \cdot \phi(x-k) \cdot \psi(y-l)$$

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$$\beta_{HL}^{\text{high}} = \sum_{k,j} \varphi F_i \cdot \psi(x-k) \cdot \phi(y-l)$$
$$\beta_{HH}^{\text{high}} = \sum_{k,j} \varphi F_i \cdot \psi(x-k) \cdot \psi(y-l) \tag{6}$$

where k and j represent the overall index, which is used to translate the function to ensure that the function covers every point of the image, and x and y represent the axis of the space. φ and ψ represent the scale function and wavelet function, respectively, as shown in the following equation:

$$\phi(t) = \sqrt{2}\sum_k h[k]\phi(2t-k)$$
$$\psi(t) = \sqrt{2}\sum_k g[k]\phi(2t-k). \tag{7}$$

The edge method then performs weighted fusion on the edges in the frequency domain and the spatial domain by using the Sigmoid function (∂) to perform pixel multiplication on θ^f with strong edge perception and weak edge β^f with complex structure, as shown in (8). This combined operation can suppress the noise generated by a single algorithm and enhance and filter the overall edge structure

$$\beta_{LH}^{\text{high}'} = \partial\left(\theta_{LH}^f\right) \times \beta_{LH}^{\text{high}} + \beta_{LH}^{\text{high}}$$
$$\beta_{HL}^{\text{high}'} = \partial\left(\theta_{HL}^f\right) \times \beta_{HL}^{\text{high}} + \beta_{HL}^{\text{high}}$$
$$\beta_{HH}^{\text{high}'} = \partial\left(\theta_{HH}^f\right) \times \beta_{HH}^{\text{high}} + \beta_{HH}^{\text{high}}. \tag{8}$$

In the low-frequency part, this article designs a change selector normalization algorithm specifically for RS change detection tasks. It uses standard deviation normalization and mean filtering methods to effectively distinguish the impact of illumination changes on image contours, and reduces the impact of nonstationary changes through dual-time normalization and equalization. Specifically, the algorithm calculates the shared standard deviation $\beta_{\text{std}}^{\text{fus}} \in \mathbb{R}^{B\times C\times H\times W}$ by channel splicing dual-phase features. And the feature average value η along the channel dimension in a single moment feature is shown in the following equation:

$$\eta_{t1} = \frac{1}{C}\sum_{c=1}^{c} \beta_{t1}^{\text{low}}. \tag{9}$$

Elements in θ^{low} that are less than or equal to the mean are set to 0, and elements greater than the mean are retained. This step helps filter out the general brightness changes caused by illumination changes and emphasizes the structural focus on the contours of real objects. Each element is then normalized according to the count and standard deviation $\beta_{\text{std}}^{\text{fus}}$ of its nonzero elements, as shown in the following equation, where (i, j) represents the position representation along the channel dimension, and Δ is a small constant used to prevent division by zero errors:

$$\text{filter}(i,j) = \begin{cases} \beta_{t1}^{\text{low}}(i,j), & \text{if } \beta_{t1}^{\text{low}}(i,j) > \eta_{t1}(i,j) \\ 0, & \text{otherwise} \end{cases}$$
$$\beta_{\text{std}}^{\text{fus}} = \text{std}\left(\left[\beta_{t1}^{\text{low}}, \beta_{t2}^{\text{low}}\right]\right)$$
$$\text{norm}_{t1} = \frac{\beta_{t1}^{\text{low}} - \frac{\sum \text{filter}(i,j)}{\sum_{i,j}(\text{filter}(i,j)>0)}}{\beta_{\text{std}}^{\text{fus}} + \Delta}. \tag{10}$$

The algorithm uses the Sigmoid function to weight the result of biphase normalization and outputs the final low-frequency contour information $\beta_{ti}^{\text{low}'}$, as shown in the following equation:

$$\beta_{t1}^{\text{low}'} = \partial(\text{norm}_{t1}) \times \beta_{t1}^{\text{low}} + \beta_{t1}^{\text{low}}$$
$$\beta_{t2}^{\text{low}'} = \partial(\text{norm}_{t2}) \times \beta_{t2}^{\text{low}} + \beta_{t2}^{\text{low}}. \tag{11}$$

We perform inverse wavelet transform on the refined high-frequency edge feature $\beta_{t1}^{\text{high}'}$ and low-frequency contour feature $\beta_{t1}^{\text{low}'}$ to increase the dimension and integrate them to obtain $\phi_{\text{fus}} \in \mathbb{R}^{B\times C\times H\times W}$. In order to further distinguish the structural information and stabilize the connection between the edge and contour of the object, this article designs a method of combining graph convolution and local convolution. Specifically, we select graph convolution of nine adjacent nodes to capture the irregular structure in the global range, and select 3×3 local convolution groups to capture the regular structure in the local range, as shown in the following equation, where $\hat{A}$ is the adjacency matrix, representing the structural relationship of the graph, and W and b represent the weights:

$$\phi_{\text{graph}} = \text{GraphConv}(\phi_{\text{fus}}) = \hat{A}_9\phi_{\text{fus}}W + b$$
$$\phi_{\text{local}} = \text{Conv}_{3\times 3}(\phi_{\text{fus}}). \tag{12}$$

At the same time, this article uses a spatial offset field for local regular structures to regulate the pixel positions of global irregular structures and maintain stable structural feature aggregation. The offset field converts the offset channel generated by ϕ_{local} into horizontal and vertical offsets $\phi_{\text{low}}^{\text{offstets}} \in \mathbb{R}^{B\times C\times H\times W}$ through convolution operations, then defines a learnable basic grid coordinate W_{coords}, and offsets and normalizes the sampling coordinate D, as shown in the following equation, where $u \in [0, W-1]$, $v \in [0, H-1]$:

$$W_{\text{coords}} = \begin{bmatrix} u + 0.5 \\ v + 0.5 \end{bmatrix}$$
$$W_{\text{normal}} = \frac{2 \cdot \left(W_{\text{coords}} + \phi_{\text{local}}^{\text{offstets}}\right)}{\begin{bmatrix} W \\ H \end{bmatrix}} - 1. \tag{13}$$

Feature ϕ_{graph} is sampled by bilinear interpolation on the offset coordinate W_{normal} to obtain $\phi e_i \in \mathbb{R}^{B\times C\times H\times W}$, thereby achieving local adaptive sampling and global irregular structure alignment to maintain the overall consistency of the image.

D. Diff-Enhance

We designed the Diff-Enhance module as a change detection method for downstream tasks, as shown in Fig. 1. First, we use cosine similarity and channel dimension aggregation on the channel dimension for the biphase features ϕe_1 and ϕe_2 to measure the difference between the two tensors in the feature space, and use the Sigmoid function to compress the cosine similarity result to the range of [0, 1] to obtain the weight $\phi_{\cos} \in \mathbb{R}^{B\times 1\times H\times W}$, as shown in the following equation:

$$\phi_{\cos} = \partial\left(1 - \frac{\phi e_1 - \phi e_2}{\|\phi e_1\|\|\phi e_2\|}\right) \cdot \text{insqueeze}(1). \tag{14}$$

The cosine weights are used to redistribute the biphase features in order to emphasize the change regions in the two features.

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The features are then fused by channel superposition and absolute value difference, as shown in the following equation:

$$\begin{aligned} \phi e_{\text{cat}} &= \text{Concate}(\phi e_1, \phi e_2) \\ \phi e_{\text{sub}} &= |\phi e_1 - \phi e_2| \\ \phi_{\text{fus}} &= \phi e_{\text{cat}} + \phi e_{\text{sub}}. \end{aligned} \tag{15}$$

The final difference feature ϕ_{fus} undergoes two layers of 3×3 convolution for simple feature enhancement. In this article, the difference features of each scale are aligned in channels and scales through deconvolution, and decoded and fused by channel superposition.

E. Loss Function

This article selects cross entropy as the loss function, which has been widely used in the field of RS change detection and has shown excellent performance. Cross-entropy effectively improves the model's ability to identify minority categories by optimizing the probability distribution of the model output, while accelerating the convergence of the training process and further improving the accuracy and confidence of the prediction results. In addition, this article selects the last three scales for loss optimization and combines the deep supervision network to constrain the model, thereby significantly enhancing the detection effect of the model. As shown in the following equation:

$$L = \frac{1}{H_0 \times W_0} \sum_{h=1,w=1}^{H,W} l(P_{hw}, Y_{hw}). \tag{16}$$

We use Loss_1, Loss_2, and Loss_3 to represent different loss values. After multiple experiments, we select the weights that best suit the model's adaptability for adjustment, as shown in the following equation:

$$L = \text{Loss}_1 + 0.5\text{Loss}_2 + 0.2\text{Loss}_3. \tag{17}$$

IV. Experiments

A. Datasets

We verify the superiority of GSSR-Net in RS change detection tasks from multiple aspects through three datasets, as follows.

The LEVIR-CD [40] dataset is designed for change detection in RS images, focusing on identifying changes in buildings at different time points. The dataset contains 637 pairs of high-resolution (0.5 m/pixel) images, recording surface changes in multiple cities in TX, USA, from 5 to 14 years, covering a variety of building types such as villas and apartments. Image annotation is completed by RS image interpretation experts and double-reviewed to ensure annotation quality. The dataset is cropped to obtain nonoverlapping image blocks of 256×256 pixels and divided into training set, validation set, and test set in a ratio of 2:7:1.

The WHU-CD [41] building detection dataset was developed by the Center for Geospatial Information Technology of Wuhan University and is dedicated to detecting and locating buildings from high-resolution RS images. The dataset contains a large number of accurately annotated building boundaries and is widely used in tasks such as surface change detection, image classification, and target detection. The dataset includes high-resolution images of multiple urban areas with a resolution of up to 0.5 m. The images are cropped into small blocks of 256×256 pixels and randomly divided into 6096 training sets, 762 validation sets, and 762 test sets, which are suitable for complex deep learning model training and evaluation.

The GZ-CD [42] dataset focuses on change detection in RS images under seasonal changes, covering the geographical area of the suburbs of Guangzhou, China, from 2006 to 2019. The dataset contains 19 pairs of high-resolution (0.55 m/pixel) images, normalized and cropped to a size of 256×256 pixels. The dataset is divided into 2834 training images, 400 validation images, and 325 test images, which are suitable for developing and evaluating change detection algorithms.

FC-Siam-Conc: An RS image change detection method based on a fully convolutional twin network, using a Siamese network architecture, specifically designed to process dual-temporal RS image pairs. The network detects changes in the same area at different time points by calculating the difference between image pairs at two moments and is particularly suitable for time-series data analysis.

B. Implementation Details and Evaluation Metrics

1) Implementation Details: We trained and tested the model on an NVIDIA RTX TITAN, using Python 3.9 and PyTorch 11.8, a learning rate of $5e^{-}4$, an optimizer of adamw, a weight decay of $1e^{-}2$, and a network batch size of 8.

2) Evaluation Indicators: In order to prove the effectiveness and advancement of GSSR-Net, this article selects five indicators, $F1$, IoU, Precision, Recall, and OA, for comparison. The $F1$ score is the harmonic mean of precision and recall, which takes precision and recall into consideration. IoU is used to evaluate the overlap between two regions and is particularly commonly used in target detection and semantic segmentation tasks. Precision measures the proportion of samples predicted by the model to be positive examples that are actually positive examples. Recall measures the proportion of actual positive examples that are correctly identified as positive examples by the model. Among them, the most critical indicators are $F1$ and IoU, and both indicators directly represent the performance of the model. The higher the parameters of the five indicators, the stronger the superiority of the model. We provide specific calculation formulas for the five indicators, as shown in the following:

$$\begin{aligned} F1 &= 2\frac{\text{Precision} - \text{Recall}}{\text{Precision} + \text{Recall}} \\ \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\ \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \\ \text{OA} &= \frac{\text{TP} + \text{PN}}{\text{TP} + \text{FN} + \text{FP}}. \end{aligned} \tag{18}$$

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C. Contrast Experiment

In order to be able to demonstrate the advantages of the methods proposed in this article, nine representative networks for RS change detection in recent years were selected in a unified experimental environment. Comparative experiments were conducted using six CNN based methods: FC-Siam-Conc [43], IFNet [44], SNUNet [45], AERNet [46], DMINet [47], and SEIFNet [48]; three Transformer-based methods BIT [21], ChangFormer [20], Changebind [49]; and one mamba-based method, RSmamba [50].

1) *IFNet:* A deep learning model that combines multiscale convolutional features for RS image change detection. The model can fully capture the detailed changes in the image, especially the change information at different scales, by fusing multilevel features. IFNet enhances the perception of subtle changes by integrating features at different levels, thereby improving the change detection effect in complex scenes.
2) *SNUNet:* A neural network with a tight information transfer mechanism that successfully retains important spatial positioning information between the encoder and decoder and within the decoder through an efficient information transfer strategy. Features at different semantic levels are optimized through a deep supervision mechanism, which effectively improves the classification effect in RS image change detection, especially in the case of complex backgrounds and fine-grained changes.
3) *BIT:* An innovative bitemporal image Transformer model that uses a space–time modeling technique based on a small number of semantic tags to effectively capture contextual information in RS images. It performs fine processing of image features to better understand the temporal relationship between images and improve the accuracy of change detection. This method is particularly suitable for processing RS data with long time spans.
4) *ChangFormer:* A Transformer-based twin network for change detection in RS images. Different from traditional CNN methods, it uses a hierarchical Transformer encoder and a MLP decoder to capture multiscale information and long-distance details of images.
5) *DMINet:* By integrating the self-attention mechanism and the cross-attention mechanism, it designs a temporal joint attention module that can promote information interaction between dual-temporal RS images. The network effectively improves the accuracy of change detection through the extraction of difference features and multilevel feature alignment, especially in suppressing task-irrelevant interference and capturing small changes.
6) *AERNet:* An attention-guided edge refinement network that specifically solves the problems of building changes, insufficient feature extraction, sample imbalance, and inaccurate boundary identification in RS image change detection. By performing feature fusion at multiple scales, it integrates multilevel contextual information through a global context feature aggregation module, thereby significantly improving the accuracy of building change detection.
7) *Changebind:* Based on the Siamese network architecture, it focuses on capturing semantic changes in dual-temporal RS images. By combining local and global features, the change encoder can capture both small- and large-scale changes from multiscale features and accurately estimate the change area. This method significantly improves the network's change detection capabilities in complex environments, especially when dealing with scenes with subtle changes.
8) *Rsmamba:* A model for change detection in RS images that uses a global modeling method with linear complexity and successfully solves the problem of context information loss. By introducing an omnidirectional selective scanning module, Rsmamba can capture spatial features in RS images from multiple directions.
9) *SEIFNet:* The Siamese hierarchical backbone network is used to extract multilayer feature maps, and the spatiotemporal difference enhancement modules (ST-DEMs) and adaptive context fusion module (ACFM) are introduced to effectively improve the feature representation ability of change detection. This article also designs a refinement module and a summation-based prediction head to enhance the boundaries and integrity of the detection results.
10) *LEVIR Dataset:* We conducted comparative experiments with GSSR-Net and nine advanced RS change detection methods. The experiments verified the superiority of this method through indicator comparison and visualization result comparison. After a large number of experiments, the indicator comparison results are shown in Table I. All the method indicators are shown in the table, with red as the best indicator and blue as the second-best indicator. Specifically, the key indicators $F1$ and IoU of GSSR-Net ranked first with 91.75%/84.87% points, respectively, and other indicators also ranked among the top with excellent points. The results of the visualization comparison experiment are shown in Fig. 4. This article selected six groups of comparison pictures and used four colors to distinguish different detection results. The specific meanings of the colors are as follows: black represents true negative classes (TN, i.e., samples correctly classified as negative classes), white represents true positive classes (TP, i.e., samples correctly classified as positive classes), green is used to represent false negatives (FN, i.e., samples that are actually positive but are mistakenly classified as negative classes), and red represents false positives (FP, i.e., samples that are actually negative but are mistakenly classified as positive classes). We can clearly observe that the overall structure of GSSR-Net is more complete and closer to the ground truth (GT) image in the six sets of visualization results. Large ground objects are often blocked by other objects or light, which affects the internal contour of the detection. Fig. 4(a)–(c) shows the results of large-scale target change detection in three sets of images. Compared with the other nine methods, which have large-scale

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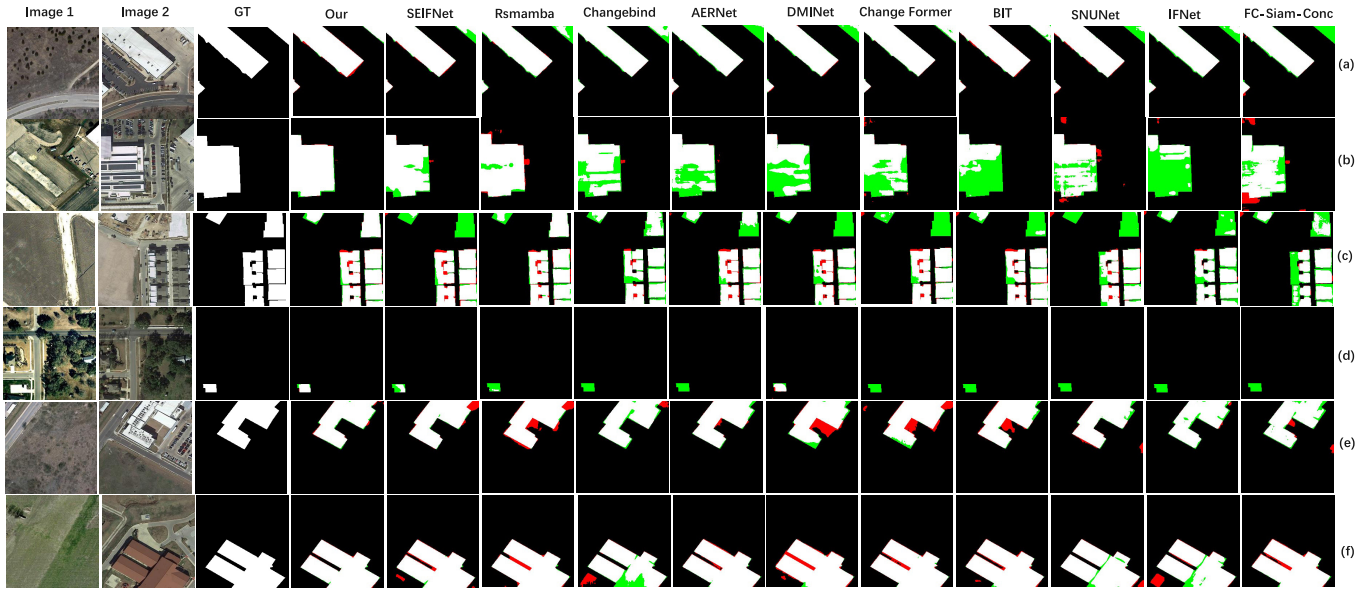


Fig. 4. We compare GSSR-Net with other models in the LEVIR-CD dataset for difference, and use colors such as red and green to interpret the changes, so as to more intuitively reflect the differences between images of different models and real images, in which white represents TPs, black represents TNs, red represents FPs, and green represents FNs. We chose (a)–(f) photographs to show the visual comparison results of different images in different models.

missed detections within the contour, our method benefits from the effective processing of low-frequency contours. GSSR-Net can perfectly identify the internal contours of large-scale targets. Small targets are often similar to or overlap with buildings, roads, trees, etc., in the environment, so their boundaries are blurred and difficult to distinguish from the background. Fig. 4(d) shows a set of visualization results for small target detection. The comparison methods generally have difficulty distinguishing their ground object structures when facing the change detection of such small targets. Our method can provide the model with sufficient semantic information to distinguish the change detection of small targets by combining local convolution with graph convolution for the connection between pixel nodes. Fig. 4(c), (e), and (f) shows three sets of contrast images with significant edge structures. These structural information often presents nonlinear and irregular shapes. Conventional methods are difficult to accurately extract complex edges. This article adopts a new idea of combining spatial strong edges and high-frequency weak edges for edges. We observe that the GSSR-Net method is significantly better than other methods.

11) *WHU Dataset:* We conduct extensive experiments on the WHU dataset under the same environment. The experimental index results are shown in Table II. Among them, the $F1$ and IoU of GSSR-Net are 93.07%/87.03% points higher than the second place Changebind by 0.76%/1.31%. OA, Precision, and Recall are all ranked in the forefront with higher indicators. In order to further highlight the superiority of the WHU dataset, we selected six groups of visual comparison images for experiments, as shown in Fig. 5. RS changing images often appear with a large number of irregular ground objects. How to capture the correct changing features among many feature structures is a problem

TABLE I

RESULTS OF INDICATORS ON THE LEVIR-CD DATASET FOR EACH COMPARISON METHOD

Method	LEVIR-CD				
	Precision	**Recall**	**F1**	**IoU**	**OA**
FC-Siam-Conc	91.99	76.77	83.69	71.96	98.49
IFNet	94.02	82.93	88.13	78.77	98.87
SNUNet	89.18	87.17	88.16	78.83	98.82
BIT	89.24	89.37	89.31	80.68	98.92
ChangeFormer	92.05	88.80	90.4	82.48	99.04
DMINet	92.52	89.95	90.71	82.99	99.07
AERNet	89.97	91.59	90.78	83.11	99.07
Changebind	94.93	88.14	91.41	84.18	99.16
Rsmamba	92.68	89.19	90.90	83.33	99.09
SEIFNet	92.45	89.51	90.96	83.42	98.42
GSSR-Net	92.55	90.95	91.75	84.87	99.17

TABLE II

RESULTS OF INDICATORS ON THE WHU-CD DATASET FOR EACH COMPARISON METHOD

Method	WHU-CD				
	Precision	**Recall**	**F1**	**IoU**	**OA**
FC-Siam-Conc	91.99	76.77	83.69	71.96	98.49
IFNet	96.91	73.19	83.40	71.52	98.83
SNUNet	91.34	85.53	88.34	79.11	98.71
BIT	86.64	81.48	83.98	72.39	98.92
ChangeFormer	90.50	79.61	84.51	73.18	98.59
DMINet	92.65	90.35	91.49	84.31	99.19
AERNet	92.47	91.89	92.18	85.49	99.25
Changebind	94.81	89.94	92.31	85.72	99.28
Rsmamba	91.34	91.57	91.45	84.26	99.17
SEIFNet	88.65	86.83	87.73	78.14	98.83
GSSR-Net	95.00	91.21	93.07	87.03	99.35

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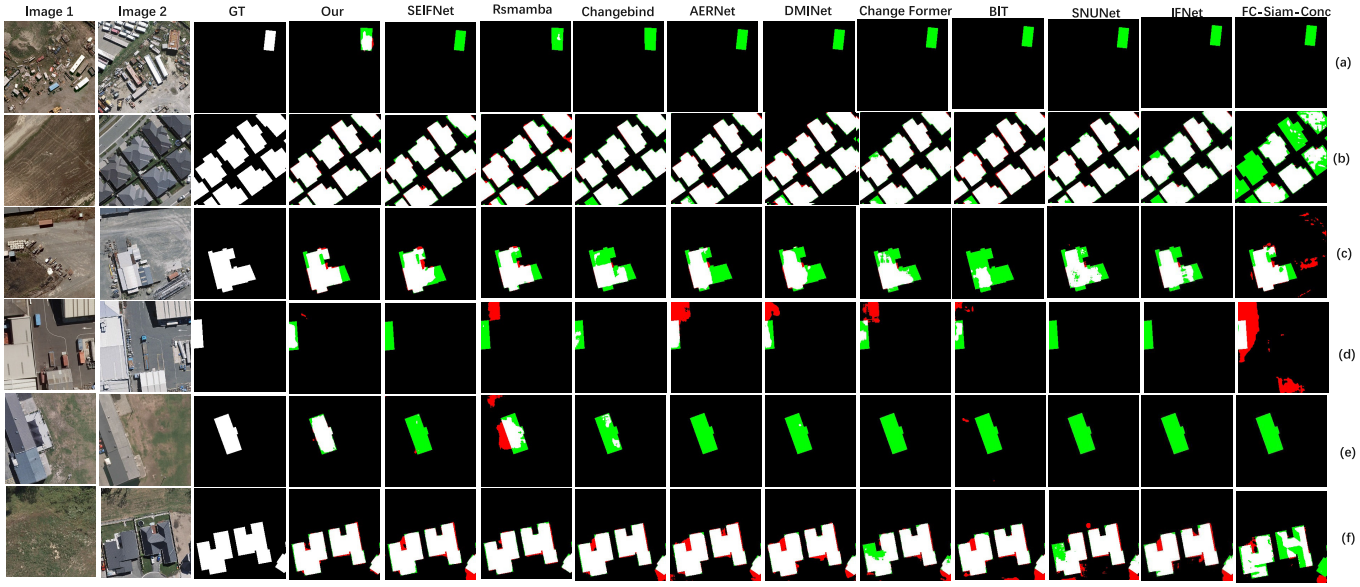


Fig. 5. We compare GSSR-Net with other models in the WHU-CD dataset for difference, and use colors such as red and green to interpret the changes, so as to more intuitively reflect the differences between images of different models and real images, in which white represents TPs, black represents TNs, red represents FPs, and green represents FNs. We chose (a)–(f) photographs to show the visual comparison results of different images in different models.

faced by all methods. Fig. 5(a) shows a set of RS images with complex environments. We can observe that all nine comparison experiments have missed detections. GSSR-Net benefits from being able to distinguish target structures, thereby overcoming irregular terrain. Physical information is used to accurately locate changing targets. Fig. 5(b), (c), and (f) shows three sets of contrast images with obvious edge structures. We can significantly observe that GSSR-Net's recognition of edge contour structures is much better than other comparison methods. Fig. 5(d) and (e) shows two sets of changing targets in closely linked areas. This situation greatly tests the network's control over the structure. Our network uses the LCFE method to greatly control nodes and nodes. The correlation between them is not only better than other methods in edge structure, but also better than other methods in anti-interference ability.

12) *GZ Dataset:* We selected a dataset with multiple nonstationary factors such as illumination differences and atmospheric factors for experimental comparison. The experiment on the GZ dataset can verify the anti-interference ability of the model for RS change detection. Table III shows the index data of nine methods and GSSR-Net. We can observe that the $F1$ and IoU of our method are 86.26%/75.84% higher than the second place by 0.96%/1.4%. Fig. 6 shows six sets of comparison images of the visualization experiment, which more intuitively demonstrates the superiority of GSSR-Net. Fig. 6(a) and (b) shows two sets of comparison images with atmospheric change interference. We can observe that the dual-phase images have obvious nonstationary change factors. GSSR-Net can effectively resist the interference of such noise through interactive coding structure, while other methods have large-scale missed detection. Fig. 6(c)–(e) shows three sets of comparison images with obvious structures. GSSR-Net uses the normalization algorithm of RS change detection to deal with the differences caused by nonstationary changes. We can observe that the experimental results of this method are far better than other methods. Fig. 6(f) shows a set of contrast images of small target structures. Our network, thanks to its advantage in low-frequency interactivity, can overcome the nonstationary interference caused by atmospheric changes and effectively capture change information that is difficult to obtain by other methods.

TABLE III
RESULTS OF INDICATORS ON THE GZ-CD DATASET FOR EACH COMPARISON METHOD

Method	GZ-CD				
	Precision	**Recall**	**F1**	**IoU**	**OA**
FC-Siam-Conc	80.59	72.34	76.24	61.61	95.68
IFNet	92.19	74.08	82.15	69.71	96.92
SNUNet	84.25	81.82	84.25	72.79	97.07
BIT	82.40	78.18	80.23	66.99	96.31
ChangeFormer	84.59	65.23	73.66	58.30	95.53
DMINet	87.92	76.79	81.98	69.46	96.77
AERNet	88.06	81.07	84.42	73.03	97.13
Changebind	87.81	82.34	84.95	73.89	97.21
Rsmamba	86.63	81.44	83.86	72.21	96.99
SEIFNet	90.07	79.95	84.71	73.47	97.23
GSSR-Net	91.19	81.83	86.26	75.84	97.50

In order to further observe the details in the experiment, we selected six groups of visual images from the comparison images of the three datasets for enlargement and comparison. As shown in Fig. 7, our method is better than the other nine methods in controlling the detailed structure. The structure

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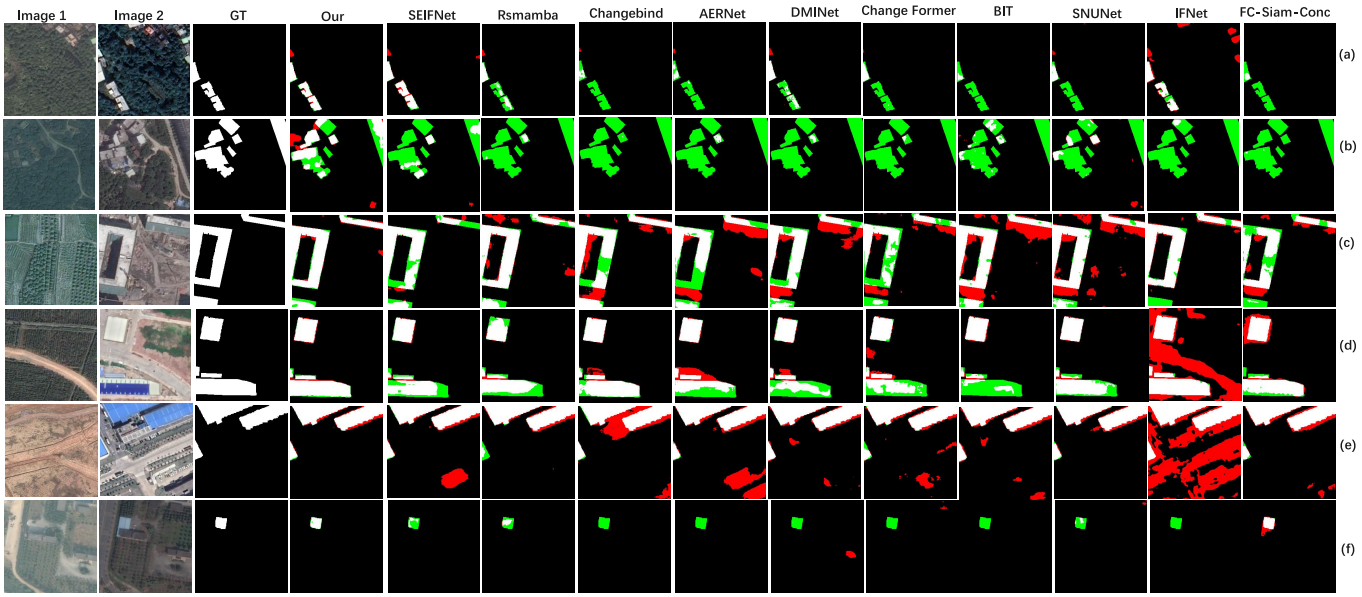


Fig. 6. Differential comparisons of all models were performed in the GZ-CD dataset, and colors such as red and green were used to explain the variations in order to better visualize the differences between different model images and the real images, where white denotes the TPs, black denotes the TNs, red denotes the FPs, and green denotes the FNs, and we chose (a)–(f) images to show the results of the visual comparisons of different images in different models.

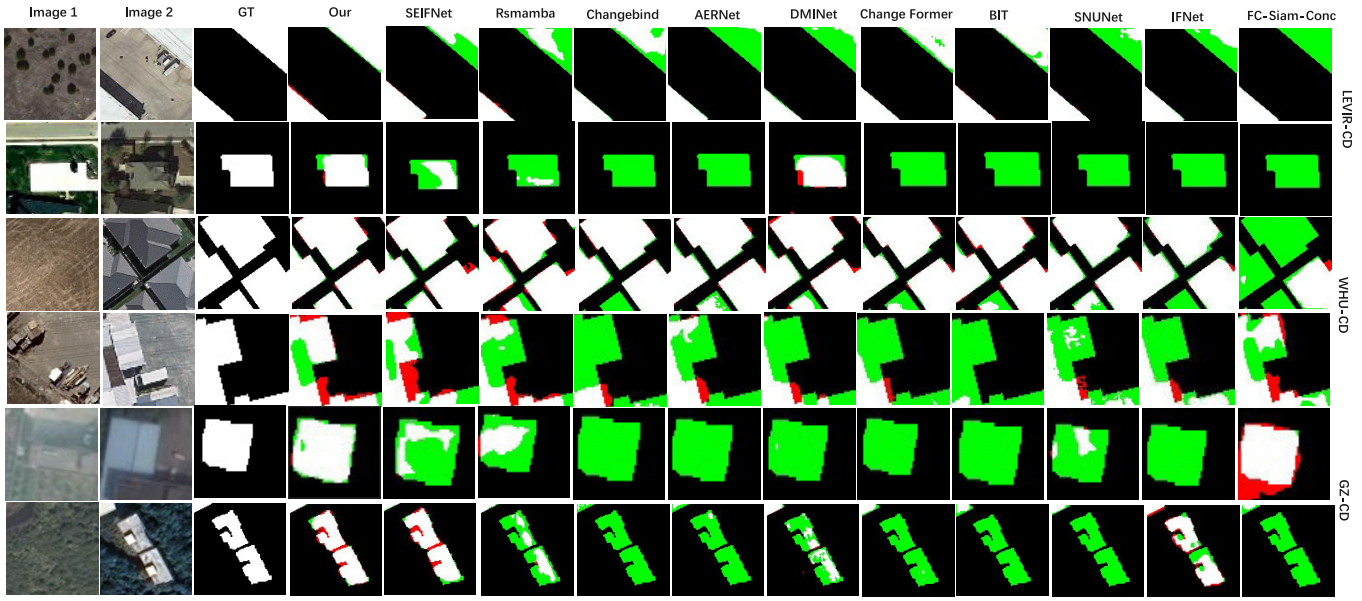


Fig. 7. Comparison of different image details in three datasets.

is more excellent. Efficiency calculation: in order to compare the efficiency of all methods, we use two indicators of computational complexity for scientific comparison, as shown in Table IV. Under the same experimental environment, our method has a Param of 27.3 and FLOPs of 15.6, which is at the forefront of all the compared models. Although the lightweight FC-Siam-Conc, BIT, and DMINet have a slight advantage over GSSR-Net, they are far behind our method in the final indicators and visualization experimental results, and these methods cannot be applied in practice. Other high-consumption network models are lower than our method in both indicators and efficiency, and their experimental results are not ideal. In summary, GSSR-Net achieves better results while maintaining high efficiency.

D. Ablation Experiment

To further demonstrate the importance of all modules in GSSR-Net, we conduct ablation experiments on the two main modules, DualTime Mamba and LCFE.

DualTime Mamba not only has the advantage of being lightweight, but also uses the unique DualTime SSM's dual-phase RS change detection interactive mechanism to overcome the influence of nonstationary factors. In order to prove the superiority of the DualTime Mamba coding

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TABLE IV
COMPLEXITY AND THE NUMBER OF PARAMETERS OF THE THREE DATASETS

Method	Complexity		LEVIR-CD		WHU-CD		GZ-CD	
	Param(M)	**FLOPs(G)**	**F1**	**IoU**	**F1**	**IoU**	**F1**	**IoU**
FC-Siam-Conc	1.55	5.32	83.69	71.96	83.69	71.96	76.24	61.61
IFNet	50.71	82.35	88.13	78.77	83.40	71.52	82.15	69.71
SNUNet	12.03	54.88	88.16	78.83	83.50	71.67	84.25	72.79
BIT	3.55	10.59	89.31	80.68	83.98	72.39	80.23	66.99
ChangeFormer	41.03	373.43	90.40	82.48	84.51	73.18	73.66	58.3
DIMNet	6.24	14.55	90.71	82.99	91.49	84.31	81.98	69.46
AERNet	25.36	12.82	90.78	83.11	92.18	85.49	84.42	73.03
Changebind	153.70	80.29	91.41	84.18	92.31	85.72	84.95	73.89
Rsmamba	27.90	185.20	90.90	83.33	91.45	84.26	83.86	72.21
SEIFNet	27.91	8.37	90.96	83.42	87.73	78.14	84.71	73.47
GSSR-Net	12.70	19.50	91.75	84.87	93.07	87.30	86.26	75.84

TABLE V
REPLACEMENT EXPERIMENT FOR ENCODERS

No.	Backbone	Complexity		LEVIR-CD		WHU-CD		GZ-CD	
		Param(M)	**FLOPs(G)**	**F1**	**IoU**	**F1**	**IoU**	**F1**	**IoU**
1	Resnet34	22.95	45.87	90.88	83.28	91.78	84.75	84.73	73.51
2	Pvt-b2	20.56	40.83	91.21	83.85	92.18	85.50	85.01	73.94
3	Vmamba	32.74	77.04	91.34	84.06	92.39	84.91	84.72	73.50

TABLE VI
DUALTIME MAMBA WAS SUBJECTED TO STRUCTURAL EXPERIMENTS ON THREE DATASETS. EACH MODULE WAS ABLATED AND THE ABLATION $F1$ AND IOU VALUES WERE CALCULATED

No.	DT Mamba	LEVIR-CD		WHU-CD		GZ-CD	
		F1	**IoU**	**F1**	**IoU**	**F1**	**IoU**
1	V0	91.53	84.39	92.7	86.39	85.39	74.51
2	V1	91.56	84.44	92.85	86.65	85.78	75.06
3	V2	91.65	84.59	92.82	86.61	85.64	74.88
4	V3	91.64	84.58	92.73	86.45	85.84	75.19

TABLE VII
LCFE WAS SUBJECTED TO STRUCTURAL EXPERIMENTS ON THREE DATASETS. EACH MODULE WAS ABLATED AND THE ABLATION $F1$ AND IOU VALUES WERE CALCULATED

No.	LCFE	LEVIR-CD		WHU-CD		GZ-CD	
		F1	**IoU**	**F1**	**IoU**	**F1**	**IoU**
1	V0	91.38	84.12	92.29	85.68	85.08	74.04
2	V1	91.58	84.48	92.67	86.34	85.4	74.52
3	V2	91.62	84.54	92.8	86.56	85.85	75.21
4	V3	91.6	84.50	92.65	86.3	85.63	74.87
5	V4	91.68	84.65	92.93	86.81	85.72	75.01

structure in this article, we selected three popular coding structures, CNN, Transformer, and Mamba, for comparison under non-pretraining conditions in the same environment. According to Table V, after conducting a large number of experiments and taking the average, the indicators of all methods are significantly lower than the coding structure we designed. These metrics demonstrate the uniqueness of our method for the change detection task and the feasibility of being broadly applicable to other vision tasks.

DualTime Mamba uses the SSM-Base structure in the shallow scale and the interactive SSM-Dual structure in the deep scale. This combination allows the network to not only fully learn the feature information of the current time point in the first two stages, but also learn the information of different time points in the subsequent two stages. To further prove the effectiveness and rigor of this combination, we arrange the SSM structures on the four scales in four combinations. As shown in Table VI, V0 uses SSM-Base in all four scales; V1 uses SSM-Base in the first three scales and SSM-Dual in the fourth stage; V2 uses SSM-Base in the first stage and SSM-Dual in the subsequent three scales; and V3 uses SSM-Dual in all four scales. After multiple experiments, we obtained the mean index, as shown in Table VI. Different arrangement indexes in the three datasets are all lower than the structure we designed. In order to improve the rigor of the ablation experiment, this article further proves the experimental conclusion through the visualization results of Fig. 8. We can clearly observe that in the V0 and V1 experiments, due to the inability to cope with the noise factors caused by nonstationary changes, missed detection occurs. In the V2 and V3 experiments, there were too many interactions between the dual-time matrix features, which led to false detections.

The LCFE method provides GSSR-Net with the ability to capture refined ground object structures. It enhances the edge by high frequency, the contour by low frequency, the connection between pixels by graph convolution and local

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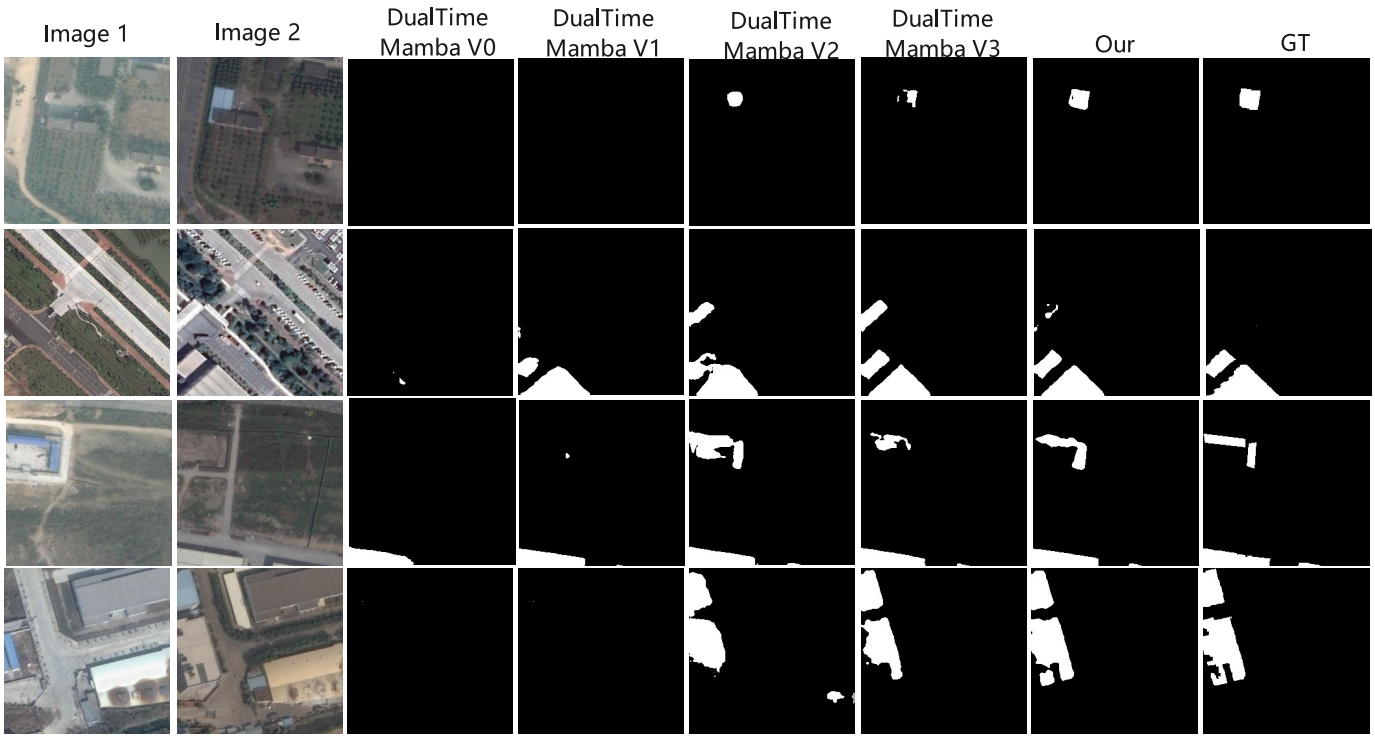


Fig. 8. In the DualTime Mamba method, SSM-Base and SSM-Dual are arranged in four combinations, where V0 is the use of SSM-Base in all four scales; V1 is the use of SSM-Base in the first three scales and SSM-Dual in the fourth stage; V2 is the use of SSM-Base in the first stage and SSM-Dual in the subsequent three scales; and V3 is the use of SSM-Dual in all four scales.

TABLE VIII
RESULTS OF HYPERPARAMETER ABLATION OF LOSS FUNCTION

No.	$Loss_1$	$Loss_2$	$Loss_3$	LEVIR-CD		WHU-CD		GZ-CD	
				F1	**IoU**	**F1**	**IoU**	**F1**	**IoU**
1	1	0.9	0.1	91.66	84.61	92.82	86.63	85.84	75.19
2	1	0.8	0.2	91.68	84.64	92.85	86.67	85.95	75.37
3	1	0.6	0.3	91.71	84.77	92.91	86.80	86.03	75.57
4	1	0.5	0.2	91.75	84.87	93.07	87.30	86.26	75.84

convolution, and the overall structurality of ground object features by spatial offset field. We designed five ablation experiments for LCFE: V0 is all methods to remove LCFE; V1 is to remove the operator structure in high frequency; V2 is to remove the normalization algorithm in low frequency; V3 uses regularized 7×7 convolution to replace irregular graph convolution; and V4 is to remove the spatial offset field linking local convolution and graph convolution. We obtained the mean index after multiple experiments, as shown in Table VII. Different permutation indexes in the three datasets are lower than the structure we designed. In order to improve the rigor of the ablation experiment, this article further proves the experimental conclusion through the visualization results of Fig. 9. We can clearly observe that in the V0 experiment, the network lacks the perception of the overall structure, resulting in large-scale false detection. In the V1 experiment, the network's perception of the edge of the ground object target is weakened, and the downstream task cannot detect changes in the edge features, resulting in the inability to distinguish the correct changes in the dual-phase feature structure. In the V2 experiment, the network lacked the ability to capture the contour structure of the ground object, and the inner contour of the target was missing in all four groups of images. In the V3 experiment, the network lacked the ability to learn irregular structures, and the detection of complex boundaries in all four groups of images was missed to varying degrees. In the V4 experiment, due to the lack of stable feature aggregation for spatial offset, the network's correction of details in the four groups of images was much lower than that of the GSSR-Net method.

In the loss function experiment, we debugged the hyperparameters of the loss weights many times. We assigned

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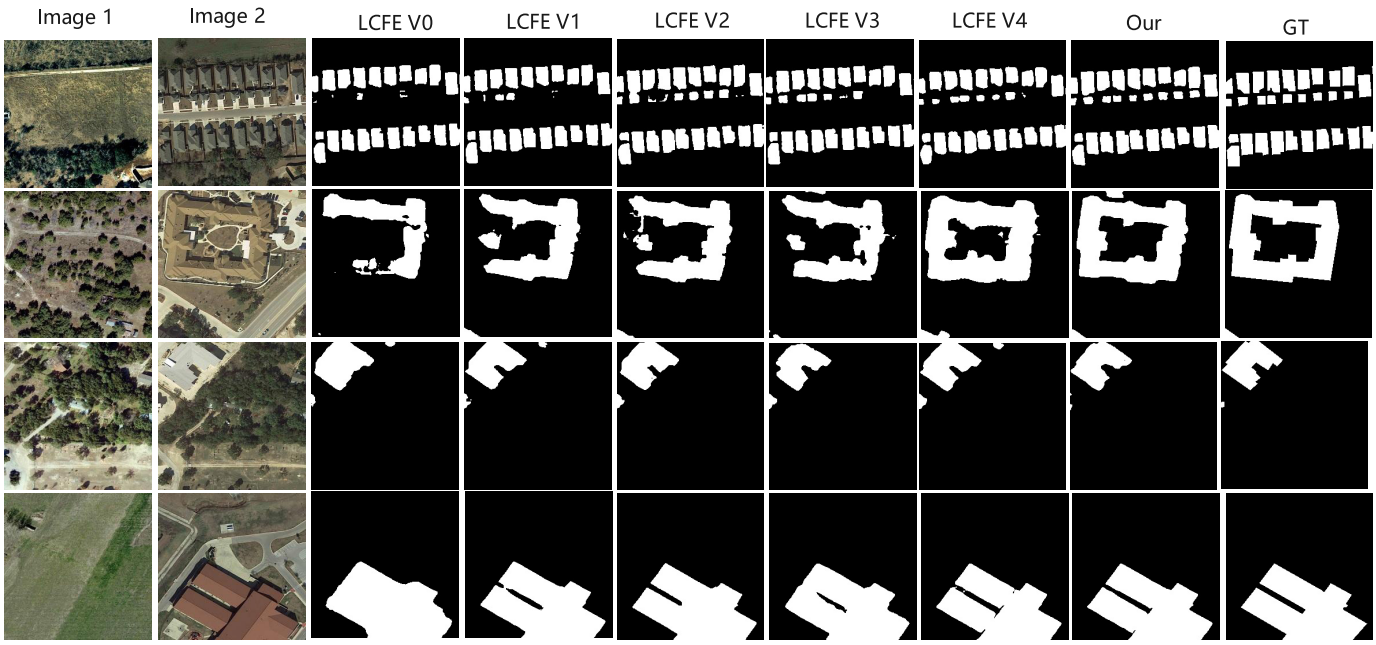


Fig. 9. Five ablation experiments were conducted on the LCFE method, where V0 is all methods for removing LCFE; V1 is for removing the operator structure in high frequencies; V2 is for removing the normalization algorithm in low frequencies; V3 uses regularized 7×7 convolution to replace the irregular graph convolution; and V4 is for removing the spatial offset field linking local convolution and graph convolution.

lower weights to information with high depth but less scale fusion; conversely, we assigned higher weights to losses that contain more information. Specifically, as shown in Table VIII, we tried multiple combinations of loss weight hyperparameters and finally selected the solution with the best performance.

V. Conclusion

This article proposes GSSR-Net to address two key challenges in RS change detection: nonstationary changes between dual-temporal image features and fuzzy distinction of ground object structure. By introducing the DualTime Mamba structure and LCFE method, GSSR-Net effectively enhances the model's perception of spatiotemporal dependence, image structure, and ground objects. The experimental results show that GSSR-Net has achieved significant performance improvements on multiple datasets, verifying its effectiveness and superiority in change detection tasks in complex RS environments. Although this method has achieved excellent results, there is still room for improvement in the generalization of model application, and its ability to address other issues with RS changes also needs to be improved. In the future, we will further optimize the network structure to achieve more efficient and cost-effective RS data processing.

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